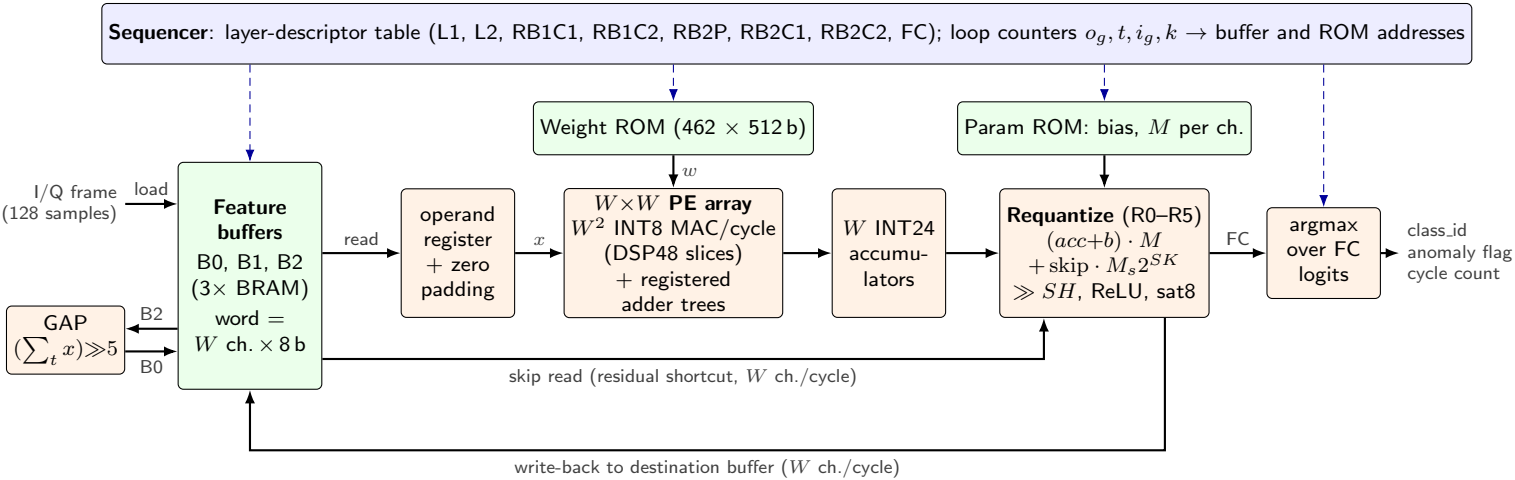

Sequencer: layer-descriptor table (L1, L2, RB1C1, RB1C2, RB2P, RB2C1, RB2C2, FC); loop counters $o_g, t, i_g, k \rightarrow$ buffer and ROM addresses
Weight ROM (462 × 512 b)
Param ROM: bias, M per ch.
w
I/Q frame (128 samples)
load
Feature buffers
B0, B1, B2
(3× BRAM)
word = W ch. × 8 b
read
operand register + zero padding
x
$W \times W$ PE array
W^2 INT8 MAC/cycle
(DSP48 slices)
+ registered adder trees
W INT24 accumulators
Requantize (R0–R5)
$(acc+b) \cdot M$
+ skip $\cdot M_s 2^{SK}$
$\gg SH$, ReLU, sat8
FC
argmax over FC logits
class_id
anomaly flag
cycle count
GAP
$(\sum_t x) \gg 5$
B2
B0
skip read (residual shortcut, W ch./cycle)
write-back to destination buffer (W ch./cycle)